PROPOSAL · FOR OFFICIAL USE — KSP / SCRB

KARNATAKA STATE POLICE · STATE CRIME RECORDS BUREAU

CRIME INTELLIGENCE & ANALYTICAL PLATFORM

DRISHTI

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Detection, Relationship-mapping, Intelligence, Spatiotemporal Hotspots & Threat-scoring Infrastructure

"Seeing the pattern before the crime — without ever inventing one."

Prepared for
State Crime Records Bureau (SCRB), Karnataka State Police

Document type
Technical & Architectural Proposal (Design only — no implementation)

Version
1.0 · 14 June 2026

★ Executive Summary

Why DRISHTI is built differently from a hackathon dashboard

DRISHTI is a Crime Intelligence & Analytical Platform designed to move the Karnataka State Police from fragmented, Excel-bound, reactive record-keeping to an integrated, spatial, relational and predictive *Strategic Intelligence Hub* — while being engineered, from the first line of architecture, around two ideas a normal hackathon project ignores: **nothing is ever fabricated, and stolen or tampered data is useless to the attacker.**

This proposal is the product of structured, parallel research across five specialist tracks — geospatial analytics, criminological network science, predictive/ML intelligence, government-grade security, and privacy-preserving assistants — with every recommended technique carrying an explicit **feasibility verdict** and citation. We deliberately included methods that are still emerging in research literature (Emerging Hot Spot Analysis, self-exciting point processes, Risk Terrain Modeling, conformal prediction, private set intersection, confidential computing) *only where they are genuinely workable at Karnataka scale*, and we explicitly rejected ideas that sound impressive but are unsafe for a police system.

7	8-layer	0	100%
Capability modules, each independently useful	Defense-in-depth security architecture	Fabricated facts — enforced in code, not policy	India-sovereign, on-prem / GovCloud deployable

The three convictions behind every design choice

1 · Zero fabrication

Human lives and liberty depend on this data. Every link, match, prediction and chatbot answer carries a *calibrated confidence and a source*. A deterministic, non-AI validator refuses any uncited claim, any unverified match, and any imputed value presented as fact. The AI is never the last line of defence for truth.

2 · Useless if stolen

"Unhackable" is a myth we refuse to sell to the police. Instead: HSM-held keys, per-subject crypto-shredding, tokenisation and confidential computing make exfiltrated data ciphertext; a hash-chained ledger makes tampering mathematically detectable; air-gapped immutable backups make ransomware *worthless as a bargaining chip*.

3 · Decision support, not decision making

Predictive policing has a documented history of bias and harm. DRISHTI predicts *places and environments*, never the dangerousness of named individuals; every model is bias-audited, explainable, uncertainty-honest, and human-in-the-loop. It informs officers; it never acts on its own.

What the platform delivers

Capability	Outcome for SCRB / KSP
Geospatial & spatiotemporal intelligence	District → police-station drill-down on GPU-rendered maps; statistically-significant hotspots; *emerging-trend alerts* that distinguish a brand-new flare-up from a chronic problem.
Network & link analysis	Connects suspects, victims, vehicles, phones and locations into one graph; reveals organised-crime structure, brokers and repeat-offender MO links invisible in isolated Excel sheets.
Predictive & anomaly intelligence	Place-based, explainable risk surfaces and significant-cluster anomaly alerts for smarter, fairer resource deployment.
Sociological correlation	Overlays urbanisation, population and socio-economic context to explain the "why behind the where" — clearly labelled as context, never individual prediction.
Private memory assistant	A grounded, citation-only AI assistant that remembers each user privately and refuses to speculate.
Citizen self-check ("MyShield")	Identity-verified citizens safely check their own records and neighbourhood safety stats — never anyone else's data.
Zero-fabrication data engine	Handles missing data with explicit "unknown" states, provenance and confidence — never silent defaults or invented values.

Not another record system

DRISHTI does **not** duplicate CCTNS or ICJS — it is the missing *intelligence layer* that sits on top of them, adding the network analysis, prediction, entity resolution and visualisation those systems were never built to do (see Part II).

How to read this document

Every technology recommendation is tagged with a verified feasibility verdict: **PRODUCTION-READY** proven and deployable today · **MATURE RESEARCH** sound, library-backed, needs engineering discipline · **EXPERIMENTAL** pilot-only, human-reviewed, never autonomous.

Contents

I The Problem

From data silos to a Strategic Intelligence Hub

The Karnataka State Police hold one of the richest crime datasets in the country — incidents, offenders, victims, locations, modus operandi. Yet the analytical value of that data is throttled by how it is stored and used today.

Today's constraint	Consequence for policing	How DRISHTI answers it
Data silos & manual Excel reporting	Each station/unit is an island; no state-wide view; analysis is slow and error-prone.	Unified, governed data platform with automated ingestion, validation and lineage.
No advanced analytics	Behavioural patterns, social ties and criminal networks remain invisible.	Graph link analysis, spatiotemporal statistics and ML — all explainable.
Information gaps at SCRB	SCRB receives fragmented inputs, weakening state-wide analysis.	SCRB becomes the hub: drill-down maps, trend alerts, network views in one console.
Reactive policing	Resources follow yesterday's crime; no systematic foresight.	Place-based, fairness-audited risk forecasting and emerging-hotspot detection.

The mandate, restated

Replace static sheets with **dynamic, spatial and relational storytelling**; connect fragmented points to reveal structure; and shift SCRB from reactive reporting to proactive, evidence-based strategy — all without compromising the integrity, privacy or legal admissibility of the underlying records.

II Landscape & Differentiation

What already exists — and the whitespace DRISHTI actually fills

A fair question for any government project: if systems already exist, why build this? The honest answer is that India's police-technology stack is strong at *recording* crime and weak at *making sense of it* — and the few analytical deployments that do exist have drawn serious legal and ethical criticism for being surveillance-heavy and person-centric. DRISHTI is deliberately positioned in that gap.

What is already in use

System	What it does well	What it does *not* do
CCTNS national · NCRB	System of record across **17,130 police stations**; ~99% of FIRs filed digitally; standardised FIR, criminal-record and investigation tracking.	A **data-entry & record system, not an intelligence platform** — no network/link analysis, no entity resolution, no prediction, no advanced visualisation. Modules are widely under-utilised; data quality is uneven.
ICJS national	Links the justice pillars — police, courts, prisons, prosecution, forensics (Phase 2, to 2025–26).	Integration plumbing, not analytics. No behavioural, spatial or relational intelligence.
Delhi CMAPS also Gurugram, Kolkata	Crime mapping & hotspot clustering from Dial-100 + ISRO satellite imagery; cut analysis time from ~15 days to ~3 minutes.	**Basic hotspot clustering only** — no emerging-trend typology, no network analysis, no entity resolution, no explainability, no privacy-preserving design.
Hyderabad "IPIH" cautionary example	A "360-degree" resident-profile hub used for person-based prediction.	**Exactly what DRISHTI refuses to be:** person-centric, surveillance-heavy, and widely criticised (LSE, SFLC, Vidhi) for privacy intrusion and discrimination risk.
Commercial LE suites Cognyte, SAS, Palantir-class	Powerful investigation and link-analysis tooling.	Costly, foreign-hosted (data-sovereignty concern), and generic — not tuned to Karnataka data, Kannada text, or Indian law (BSA / DPDP).

The whitespace DRISHTI fills

1 · Intelligence, not just records

DRISHTI is the **analytics layer on top of CCTNS / ICJS** — it integrates with them rather than duplicating them, adding exactly what they lack: network/link analysis, entity resolution, explainable prediction and relational visualisation.

3 · Fixes "linkage blindness"

The documented research gap — analysts missing connections because of poor, fragmented data — is attacked head-on with **entity resolution + NLP on FIR text + a knowledge graph**, none of which CCTNS provides.

2 · Place-based, rights-respecting alternative

Against the legal backlash facing person-based surveillance systems, DRISHTI's **place-based, explainable, privacy-preserving, auditable** design is genuinely novel in the Indian policing context — and far more defensible in court.

4 · Explainability & tamper-evidence

Existing systems are repeatedly criticised for opacity and weak data protection. DRISHTI makes **Explainable-AI, zero-fabrication and "useless-if-stolen" security** first-class — an unmet need across the board.

Positioning in one line

CCTNS asks *"what happened, and where is it recorded?"* DRISHTI answers *"what does it mean, what is connected, what is emerging, and what should we do — provably, fairly, and without inventing anything?"* It complements the national systems; it does not compete with them.

III Solution Overview & Design Principles

Seven modules on one sovereign, secure, ethically-governed core

DRISHTI is a layered platform: a governed data foundation feeds seven capability modules, all wrapped by a cross-cutting security, ethics and audit fabric. Each module is independently valuable, so the platform delivers value incrementally rather than as a risky big-bang.

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┌─ DRISHTI PLATFORM (conceptual) ─────────────────────────────────────────┐
│                                                                          │
│  PRESENTATION   Geo console · Network canvas · Predictive dashboards ·   │
│                  DRISHTI Assistant · MyShield citizen portal              │
│  ──────────────────────────────────────────────────────────────────     │
│  INTELLIGENCE   [M1] Geo/ST hotspots  [M2] Graph & link  [M3] Predict/anomaly  │
│                  [M4] Socio-economic    [M5] Assistant     [M6] MyShield        │
│                  [M7] Zero-fabrication data engine (cross-cutting)              │
│  ───────────────────────────────────────────────────────────────────    │
│  DATA           PostgreSQL/PostGIS · Graph store · Vector store · Object/WORM   │
│                  Validation · Entity resolution · Lineage · Provenance         │
│                                                                                 │
│  SECURITY & ETHICS FABRIC  Zero-Trust · HSM/keys · tamper-evident ledger ·    │
│                  bias audit · explainability · human-in-the-loop · full audit   │
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Design principles

P1 · Truth over completeness
Better to say "not known" than to guess. Confidence and provenance travel with every value.

P2 · Sovereign by default
On-prem / NIC MeghRaj GovCloud, keys in Indian HSMs. Sensitive raw data never leaves KSP's perimeter.

P3 · Open where it matters
Open-source core (PostgreSQL, MapLibre, deck.gl) to avoid lock-in; commercial tools only where they de-risk delivery.

P4 · Explainable by construction
Every match, link and score can be traced to its evidence and shown to a court or oversight body.

P5 · Human-in-the-loop
AI generates leads, never verdicts. No automated enforcement, ever.

P6 · Incremental & modular
Each module ships and proves value on its own; failure of one never cascades.

IV The Seven Capability Modules

Each technique below has been feasibility-verified for Karnataka-scale data

Module 1 — Geospatial & Spatiotemporal Hotspot Intelligence

M1 · WHERE & WHEN

District-to-station drill-down maps, statistically-honest hotspots, and the headline capability SCRB needs: alerts that tell apart a *brand-new* flare-up from a chronic problem area — because they demand opposite responses.

Methods (each chosen for a specific job, not interchangeable)

Technique	Role & why it beats the naive choice	Verdict
Getis-Ord Gi* (PySAL esda)	Primary *statistically significant* hotspot map. A raw count choropleth just shows where people live; Gi* tests significance against the local neighbourhood — defensible for resource allocation.	PRODUCTION
Local Moran's I (LISA)	Surfaces spatial *outliers* (a calm station ringed by hot ones) — signals displacement or under-reporting. Complements Gi*.	PRODUCTION
Kernel Density (KDE)	The intuitive heat layer for briefings — used as *visual* only, explicitly labelled non-inferential (no significance test).	PRODUCTION
ST-DBSCAN / HDBSCAN	Discrete event-cluster / crime-spree detection using *space and time* together. Run on time-windowed slices to stay within scale.	MATURE
Emerging Hot Spot Analysis Gi* over a space-time cube + Mann-Kendall	**The emerging-trend engine.** Classifies each location as New / Intensifying / Persistent / Diminishing / Sporadic. This is the rigorous form of your "red-zone pulsing" alert.	MATURE

Visualization & geospatial stack

MapLibre GL JS open-source base map (no vendor lock-in) deck.gl 9 GPU layers: 3D hex extrusions, animated time-slider trips, heat Uber H3 v4 hexagonal index unifying DB → analytics → render Apache ECharts & Observable Plot time-of-day & trend charts PostGIS + h3-pg Martin (Rust, fastest vector tiles) kepler.gl / pydeck analyst exploration.

The single most important design choice in M1

Standardise on **Uber H3 hexagons as the spatial unit end-to-end**. The same hex index drives PostGIS binning, Gi*/EHSA statistics and deck.gl rendering — keeping analytics, database and the "extremely good" visuals as one coherent system instead of three disconnected ones, and giving seamless district → station → street-block drill-down by changing one resolution parameter.

Honest risk flagged, not hidden

Police-station *jurisdiction boundaries* for Karnataka are rarely public and are a real data-acquisition dependency — mitigated by H3 hex aggregation, which needs no boundary files. And any forecasting layer here inherits the predictive-policing bias caveats in Part V.

Module 2 — Criminological Network & Link Analysis

M2 · WHO & HOW

This module replaces "independent silos" with a single graph of suspects, victims, vehicles, phones and locations — revealing organised-crime structure, brokers, and repeat-offender MO links that are impossible to see in isolated spreadsheets.

The governing rule for the entire module

A false association here can mean a wrongful detention. **Every link carries a calibrated probability and a provenance, never a fabricated certainty.** Mid-confidence matches go to a human-review queue; they are never auto-merged or presented as fact.

Entity resolution — the most critical component

The same offender appears as "Mahesh Kumar / Mahesh K. / M. Kumar / Magesh" across stations. Without resolution, every downstream graph is wrong — either missing real links or fabricating false ones. Your "no hallucinated links" requirement is fundamentally an entity-resolution calibration problem.

Component	Why	Verdict
Splink (Fellegi-Sunter + EM)	**Core ER engine.** Outputs a *calibrated match probability per pair* with explicit thresholds (auto-merge / review / reject) and waterfall charts explaining *why* two records matched — auditable and court-defensible. Scales DuckDB → Spark to 100M+ records.	PRODUCTION
Indic-transliteration blocking (Sentence-BERT + FAISS)	Generates candidate pairs across Kannada↔English romanisation variants, feeding Splink for the final calibrated, explainable score.	MATURE

Graph store, structure discovery & MO linkage

Component	Why	Verdict
Neo4j + GDS (sovereign alt: Apache AGE on Postgres)	Constant-time multi-hop traversal ("everyone within 3 hops of this suspect"); richest built-in graph-algorithm library.	PRODUCTION
Leiden community detection (not Louvain)	Finds candidate gangs/syndicates and *guarantees well-connected communities* — Louvain can report disconnected "gangs" that aren't actually linked (a fabricated structure).	PRODUCTION
Betweenness centrality + k-core	Identifies *brokers / kingpins* (Morselli's brokerage thesis) and the dense inner ring vs peripheral associates — presented as leads, caveated for network incompleteness.	PRODUCTION
Structured MO features + Jaccard	**Repeat-offender / serial linkage.** Encodes entry method, tools, timing, target; Jaccard correctly ignores joint-absences (ambiguous in messy FIRs). Explainable and ROC-thresholded.	PRODUCTION
GraphSAGE / GAT link prediction + SHAP/GNNExplainer	Suggests *possible unrecorded* associations — but only as ranked, explained leads. **Pilot-only, human-reviewed, barred from any automated action.**	EXPERIMENTAL

Investigator-grade visualization

`sigma.js v3 + graphology` WebGL canvas (100k+ edges) `Cytoscape.js` case / ego-network detail `Gephi` offline court exhibits `evaluate: Linkurious / ReGraph` purpose-built LE link-analysis (buy-vs-build de-risking).

Module 3 — Predictive Risk & Anomaly Intelligence

Fairness leads, because harm — not inaccuracy — is the dominant risk

Person-based predictive policing has a documented record of failure (Chicago's "Heat List", terminated 2019 as inaccurate and racially biased; PredPol feedback loops that launder over-policing into "objective" prediction). DRISHTI is built to *break* those loops: **predict places and environments, never named individuals; report uncertainty, never false precision; support decisions, never make them.**

Risk forecasting — place-based and defensible

Method	Why	Verdict
Risk Terrain Modeling (RTM) Caplan & Kennedy	**The interpretable backbone.** Forecasts risk from the *environmental features* of a place (transit nodes, ATMs, markets, vacant lots) — tells you *why* a place is risky and what to mitigate. Far more defensible against bias critiques than arrest-based prediction.	PRODUCTION
LightGBM / CatBoost + SHAP	Workhorse for tabular FIR data (area, time, type, calendar). SHAP gives exact per-prediction reasons — every score is explainable.	PRODUCTION
Hawkes / self-exciting point process Mohler et al.	Models *near-repeat victimisation* (burglary, vehicle theft, retaliatory violence cluster after a trigger). Targeted near-repeat alerts only.	MATURE
Conformal prediction (MAPIE)	**Anti-hallucination core.** Wraps every score with a coverage-guaranteed interval — a wide interval explicitly says "low confidence, gather more info" instead of a false-precise number.	MATURE
Graph WaveNet / ST-ResNet	Deep spatiotemporal accuracy layer — *Phase-2 only*, behind the interpretable spine, with SHAP + uncertainty + bias audit attached. Never a standalone oracle.	EXPERIMENTAL

Anomaly detection — layered

Isolation Forest + Local Outlier Factor fast tabular triage → SaTScan space-time scan statistics significance-tested emerging clusters → autoencoders advanced phase. Every anomaly is a *flag for human review*, with thresholds audited so they don't simply re-flag already heavily-policed areas.

Trust instrumentation (mandatory, cross-cutting)

AIF360 + Fairlearn bias audit gate (pre- & post-deploy) SHAP on every score Model Cards intended & prohibited uses MLflow registry/provenance Evidently drift monitoring DVC data versioning.

Module 4 — Sociological & Socio-Economic Correlation

M4 · THE WHY

Overlays crime with **Census of India**, urbanisation/built-up layers (WorldPop, GHSL, ISRO Bhuvan) and socio-economic indicators, analysed with **Geographically Weighted Regression** to explain the “why behind the where” and turn a risk surface into *environment-modifying* interventions.

Mandatory caveats — surfaced in the UI, not buried in docs

Ecological fallacy: area statistics say nothing about individuals living there — the platform never infers individual propensity from neighbourhood data. **Proxy bias:** poverty/literacy variables correlate with *recorded* crime partly through policing patterns; treated as analyst context, scrutinised or excluded as predictive features. **MAUP:** results vary with the chosen spatial unit — sensitivity is reported.

Module 5 — DRISHTI Assistant: the Private Memory Chatbot

M5 · ASK ANYTHING, GROUNDED

A conversational assistant that *remembers each user privately* across sessions, answers *only* from authorised data with citations, and refuses to speculate. Built so one user can never see another's conversation, and so the AI is never the last line of defence for either privacy or truth.

Dual-track model strategy

Tier 1 — Claude (`claude-opus-4-8` for hard case-reasoning, Sonnet for high-volume Q&A) for redacted / aggregate / synthetic workloads, hosted in-region. Leading instruction-following and refusal behaviour serve the anti-hallucination mandate.

Tier 2 — on-prem open model (Llama 3.x / Mistral via vLLM) behind the air gap for sensitive raw record text that cannot leave KSP's perimeter. A policy router picks the tier per request from the data-sensitivity label of the retrieved context.

Private memory architecture

Mem0 per-user/session memory scopes pgvector (→ Qdrant at scale) PostgreSQL Row-Level Security isolation enforced at the datastore, not trusted to app code or the LLM per-tenant KMS keys .

Privacy of conversations

Every memory row carries an immutable `tenant_id + user_id`; all retrieval is RLS-filtered by the authenticated identity so isolation holds even if app code has a bug. Microsoft Presidio (with custom Aadhaar/PAN/vehicle recognisers) redacts PII before any Tier-1 call and before writing memory.

Anti-hallucination guardrails

RAG with mandatory citations — answers only from retrieved authorised records, each citing source IDs; if nothing authorised is retrieved it says *"data not available"*. Defence in depth: Llama Guard 4 + Prompt Guard 2 (RAG injection defence) + NeMo Guardrails (orchestration only — vendor-flagged beta, never the sole safety layer).

The backstop

A **deterministic, non-AI validator** rejects any answer lacking a valid citation to authorised data. Fabrication is blocked in code, not by trusting the model.

Module 6 — MyShield: Citizen Self-Check

M6 · YOUR DATA, ONLY YOURS

Lets an identity-verified citizen add their own information and visualise it against the crime database to catch patterns — *without* exposing anyone else's data and without fabricating matches.

Privacy-preserving by design

Primary design: a server-side authorised query returns *only* (a) records legally the requester's own (an FIR they filed, their vehicle flagged stolen) and (b) *de-identified aggregate* neighbourhood stats ("X thefts within 1 km in 90 days" — counts only, never third-party PII). Simple, auditable, safest default.

Private Set Intersection is reserved *only* for a specific, crypto-reviewed "is my identifier on a sensitive watchlist?" check — not the primary design.

Identity, consent & anti-abuse

DigiLocker / Aadhaar eKYC gate before any personal-match result — otherwise self-check becomes a database-enumeration tool. DPDP-compliant consent + data minimisation + correction/erasure. Identity-bound rate limiting + query-pattern anomaly detection + full audit log. **The match path contains no AI** — an AI may *explain* a returned result but may never *assert* a match; "no match" is reported as "no record found", never softened or invented.

Module 7 — The Zero-Fabrication Data Engine

M7 · CROSS-CUTTING

Your hardest constraint — *handle missing data efficiently but never hallucinate, because lives are at stake* — is treated as a first-class engineering subsystem, not an afterthought. It governs how every other module reports incompleteness.

Principled missing-data handling

Classify the mechanism (**MCAR / MAR / MNAR**) — it dictates whether imputation is even valid. **MICE** (multiple imputation, quantifying uncertainty) and KNN are permitted *for aggregate analytics only*.

The hard rule (non-negotiable)

Every imputed value is flagged `is_imputed=true` with method & confidence and is **physically separated** from observed values. Imputed data may feed population-level analytics only — *never* an officer's operational view, a citizen result, or anything evidentiary, and never presented as fact.

Quality, provenance & honesty

`Great Expectations / Soda` pipeline validation `explicit UNKNOWN states` never coerce missing to `0`/`""`/a default — that is exactly how fabricated facts enter a system `OpenLineage` column-level lineage so observed vs derived vs imputed is always traceable `uncertainty visualization` shaded ranges and "imputed" badges instead of false-certain point values.

LLM behaviour: strict grounding with a mandatory "I don't know / data not available" response; the same deterministic validator from M5 rejects any answer that fills a gap without a source.

V Extended Feature Catalog

Phase 2+ capabilities that deepen value — each inheriting the platform's zero-fabrication, sovereignty & human-in-the-loop guarantees

Beyond the seven core modules, the following capabilities extend DRISHTI across the full policing value chain. Each is feasibility-tagged, fits the existing security, privacy and ethics fabric, and several directly close the gaps identified in Part II. They are sequenced as Phase 2+ enhancements (see the roadmap).

A · Smarter data intake — the highest-ROI, lowest-risk layer

Feature	Value	Verdict
NLP on FIR narratives	Auto-extract people, vehicles, weapons, locations, events & MO from free-text FIRs; auto-tag crime type; Kannada/English. *Directly fixes CCTNS data-quality & linkage-blindness.*	PRODUCTION
Vulnerability-indicator detection	Instruction-tuned LLMs flag DV / child-risk / mental-health signals in narratives so vulnerable cases are prioritised.	MATURE
Voice-to-text Dial-112 intake	Kannada ASR transcribes, triages and geo-tags emergency calls.	PRODUCTION
OCR for the paper backlog	Digitises legacy handwritten FIRs & case files into searchable text.	PRODUCTION
Smart FIR de-duplication	Detects the same incident filed at multiple stations (reuses entity resolution).	PRODUCTION
Data-completeness scorecards	Surface *where* records are missing fields — nudges quality without fabricating values.	PRODUCTION

B · Investigation acceleration

Feature	Value	Verdict
"Cases like this one" recommender	Surfaces similar / serial cases by MO + geo + time.	PRODUCTION
Grounded case-file summarisation	Summarises a dossier for a new IO — cited, never invented.	PRODUCTION
Auto timeline reconstruction	Builds a chronological timeline of a suspect / incident from scattered records.	PRODUCTION
CDR / IPDR / tower-dump analysis	Call-network link analysis, co-location, common contacts — how Indian investigations actually work; plugs into the graph module.	MATURE
ANPR / vehicle-movement linkage	Ties vehicles to incidents and reconstructs movement (privacy-gated).	PRODUCTION
Digital-evidence media analysis	Object / weapon detection & OCR in seized media — *mandatory human verification, no autonomous identification.*	MATURE

C · Proactive & preventive

Feature	Value	Verdict
Near-repeat victimisation early-warning	Place-based alerts + target-hardening advice for likely-next locations.	MATURE
Missing-persons module	Risk-tiered triage, repeat-disappearance prediction, cross-matching with unidentified bodies, structured alerts.	MATURE
Repeat-offender lifecycle tracking	Bail / parole / release alerts; links to rehabilitation.	PRODUCTION
Patrol & beat optimisation	Allocates patrols against forecasted hotspots + minimises response time.	PRODUCTION
Crime-against-women / vulnerable-cohort dashboards	Dedicated POCSO / DV / dowry tracking with helpline integration — a Karnataka priority.	PRODUCTION

D · Citizen-facing — extends MyShield

e-FIR & complaint-status tracking · public aggregated safety map + safer-route · women's safety SOS integration · multilingual citizen chatbot (grounded) — all de-identified, consent-based and DPDP-compliant.

E · Command, strategy & accountability

Feature	Value	Verdict
Executive command dashboard	State-wide real-time situational awareness for DGP / SCRB.	PRODUCTION
Auto-drafted intelligence briefings	Daily / weekly grounded, cited reports — saves analyst hours.	PRODUCTION
Oversight & bias-monitoring dashboard	Who-accessed-what + fairness metrics over time — turns the ethics framing into a visible, defensible feature.	PRODUCTION
Prosecution-support / case-readiness scoring	Evidence-completeness checklist, conviction-rate analytics, BSA §63 linkage.	PRODUCTION

F · Specialised crime verticals — high-impact for Karnataka

Feature	Value	Verdict
Cybercrime & financial-fraud network module	Mule-account tracing, phishing rings, follow-the-money graphs — Bengaluru is India's cyber-fraud epicentre; a standout differentiator.	MATURE
Organised-crime escalation early-warning	Anomaly detection on criminal-network growth.	MATURE

G · Platform & integration — the must-haves

CCTNS / ICJS integration near-mandatory for adoption · NAFIS fingerprints · Vahan / Sarathi vehicle & DL · offline / low-connectivity field mode · secure mobile officer app · multi-officer case workspaces .

Top five highest-impact additions

1. NLP on FIR narratives (unlocks data quality for everything else) · **2.** CDR / network analysis (matches real Indian investigative practice) · **3.** Cybercrime / financial-fraud module (uniquely high-value for Karnataka) · **4.** CCTNS / ICJS integration (credibility & adoption) · **5.** Oversight & bias dashboard (makes the ethics posture a sellable feature).

Handle-with-care — capable but ethically / legally loaded

Facial recognition / live CCTV matching, **social-media OSINT for unrest prediction**, and **any individual-level risk scoring** are deliberately gated: if adopted at all, strictly as human-confirmed leads under explicit legal authority and oversight — never autonomous identification, never a person-level "dangerousness" score. This preserves the place-based, rights-respecting red line that differentiates DRISHTI from the systems criticised in Part II.

Reference Implementation — a working, free demo

To prove the design is buildable, DRISHTI ships with a **runnable reference implementation**: a FastAPI application that serves all seven views, deployable **free** on Render, with a seeded database of **8,000 synthetic Karnataka crime records** (plus ~19,000 persons and ~4,000 vehicles) and large multi-format test files (CSV, JSON, NDJSON, GeoJSON, Excel). All sixteen API endpoints — hotspots, emerging trends, network & community detection, entity resolution, MO linkage, anomaly detection, place-based risk, the grounded assistant, MyShield and ingestion — are verified working end-to-end.

Same code, demo ↔ production — only the engine changes

The identical codebase runs as the *free demo* (SQLite + deterministic grounded assistant, zero API keys) or in *full production* by swapping environment variables only — PostgreSQL/PostGIS, Neo4j, Qdrant, and Claude (`claude-opus-4-8`) or self-hosted Llama. Features are identical; only the models, frameworks and infrastructure differ.

Improvements added during implementation

- **Bring-your-own-data ingestion** — an upload endpoint accepts CSV/JSON/NDJSON/GeoJSON/Excel with a column-mapping file, so KSP can connect existing datasets; it **reports missing fields rather than ever auto-filling them** (the zero-fabrication rule, enforced in code).
- **Visible model-provenance tag** — the assistant labels each answer as free-extractive or a named model, making the free-vs-paid distinction transparent to the user.
- **Human-review banding surfaced in the UI** — entity-resolution results show calibrated scores with an explicit "review" band and the evidence behind each match, never a silent auto-merge.
- **"Add-new-data, everything still works" flow** — a separate recent-incident batch lets reviewers ingest fresh data live and watch every view (maps, network, trends) update.

VI Security Architecture

Tamper-evidence, resilience, and “stolen data is useless” — honestly engineered

An honest promise, deliberately made to the police

“Unhackable” and “auto-destruct on intrusion” are marketing fictions, and the second is *dangerous*: a false trigger — or an attacker deliberately tripping it — would weaponise your own kill-switch to destroy police evidence. What DRISHTI actually delivers is the real outcome you asked for: **stolen data is cryptographically useless, tampering is mathematically detectable and reversible, and ransomware has zero leverage** — via defence-in-depth, crypto-shredding, a tamper-evident ledger, and immutable air-gapped backups.

Threat → control mapping

Threat	Attacker objective	Neutralising controls
Insider (rogue/coerced officer or admin)	Bulk-export PII, alter records	Zero-Trust + ABAC least-privilege, threshold keys (no single admin), field encryption, immutable ledger, UEBA/DLP
Ransomware	Encrypt/delete data, extort	WORM Object-Lock, 3-2-1-1-0 + air-gapped backups, EDR/Falco, segmentation
Exfiltration (breach / nation-state)	Steal & leak/sell data	HSM envelope encryption, tokenisation, crypto-shred readiness, confidential computing, DLP egress
Tampering (evidence sabotage)	Quietly alter evidence	Hash-chained Merkle ledger, RFC 3161 timestamping, BSA §63 hash certificates, WORM evidence store
Nation-state / APT	Persistent stealth, supply-chain implant	SLSA + Sigstore + SBOM, HSM-rooted PKI, air-gapped tier, SOC/SIEM, attestation

The eight layers

```
L0 GOVERNANCE     ISO 27001 ISMS · DPDP 2023/Rules 2025 · CERT-In (180-day logs,
                  6-hr report) · BSA §63 evidence law · periodic VAPT
        |
L1 IDENTITY       Zero Trust (NIST SP 800-207) PDP/PEP · FIDO2 passkeys + PKI
                  smartcards · phishing-resistant MFA · RBAC+ABAC · JIT elevation
        |
L2 NETWORK        Micro-segmentation (evidence / analytics / admin planes) ·
                  SIEM + 24x7 SOC · Wazuh HIDS+FIM · Falco runtime · UEBA/DLP
        |
L3 DATA           In transit: TLS 1.3 · At rest: AES-256 + field/column enc (PII)
                  In use: confidential computing (SEV-SNP/TDX/Nitro) for biometrics
                  Useless-if-stolen: tokenisation + per-subject crypto-shred keys
        |
L4 KEY GOV        HSM (FIPS 140-3 L3, on-prem/NIC) — master key never leaves HW
                  Envelope encryption · Vault dynamic secrets · Shamir M-of-N quorum
        |
L5 INTEGRITY      Hash-chained + Merkle append-only audit ledger (QLDB-style /
                  Trillian) · per-record SHA-256 anchored → detect + roll back
        |
L6 RESILIENCE     WORM / Object-Lock COMPLIANCE MODE (root cannot delete) ·
                  3-2-1-1-0 backups · air-gapped copy → ransomware has NO leverage
        |
L7 SUPPLY CHAIN   SLSA provenance · Sigstore/cosign signing · SBOM ·
                  SAST/DAST/SCA in CI/CD · CIS-hardened distroless containers
        |
INCIDENT KILL     Authorised crypto-shred + key revocation (NOT auto self-destruct)
                  + tamper roll-back + immutable restore
```

"Useless if stolen"

- **Crypto-shredding** — per-subject keys in the HSM; destroying the key renders data permanently unrecoverable (NIST SP 800-88 *Purge*). A provable, instant kill-switch *without* the false-positive danger of auto-wipe.
- **Tokenisation + field encryption** — an exfiltrated analytics DB contains no real identities to leak or ransom.
- **Confidential computing** — SEV-SNP / TDX / Nitro keep data encrypted even *in use*, so even a root admin or malicious hypervisor sees ciphertext (applied to the most sensitive workloads only).

"Cannot be tampered or ransomed"

- **Hash-chained Merkle ledger** — every access/edit/export is anchored; a published digest proves in log(n) that *nothing was altered or deleted*. Silent insider edits become mathematically impossible to hide.
- **Integrity hashing + roll-back** — live data is continuously compared to known-good hashes; any mismatch flags tampering and is rolled back from WORM snapshots.
- **Air-gapped immutable backups** — a clean restore always exists, so encryption-for-ransom loses all leverage.

Court admissibility & Indian compliance

`BSA §63 hash certificates` (replaced IT Act §65B; effective Jul 2024) — evidence is *born court-ready* `RFC 3161 timestamping` `PKI digital signatures` non-repudiation `CERT-In` 180-day in-country logs + 6-hour breach reporting `DPDP Act 2023 + Rules 2025` `data localisation` on-prem / NIC MeghRaj GovCloud `ISO/IEC 27001` .

VII Ethics, Fairness & Legal Governance

The safeguards that make this deployable in a democracy

For a state law-enforcement platform, the governance layer is not paperwork — it is what keeps the system lawful, accountable, and resistant to the failures that have shut down predictive-policing programmes elsewhere. These rules are written into the architecture, not bolted on.

Hard rule (written into the system)	How it is enforced
Place-based, never person-based prediction	No individual risk scores or watchlists. Models forecast environmental risk of *places*; person-level targeting is architecturally excluded.
Decision support, never decision making	No automated dispatch or enforcement. Every AI output is an explained lead for a human officer who remains accountable.
No fabricated links or matches	Calibrated probabilities + thresholds + a human-review band; a deterministic validator blocks uncited claims.
Uncertainty always shown	Conformal intervals and confidence badges; the system reports what it does *not* know.
Bias audited before & after deployment	AIF360 + Fairlearn gate across districts and strata; disparate-impact checks on patrol intensity.
Explainable & documented	SHAP on every score; Splink waterfalls; shared-MO features; Model Cards stating intended *and prohibited* uses.
Fully auditable	Every automated suggestion and every human decision/override is logged to the tamper-evident ledger.
Lawful by construction	DPDP consent & minimisation, BSA §63 admissibility, CERT-In reporting, data localisation.

One-line thesis

Build a place-based, environmentally-explained, uncertainty-honest *decision-support* system on an RTM + gradient-boosting backbone, instrumented end-to-end for bias auditing and human oversight — and treat deep models and graph-neural-network leads as monitored, human-reviewed enhancements, never as an unaccountable oracle.

VIII System Architecture & Technology Stack

Compatible, sovereign, open-core — every component verified

Master technology stack

Layer	Technologies	Posture
Spatial core	Uber H3 v4 (unifying index), PostGIS + h3-pg, Martin vector tiles	Open-source
Data platform	PostgreSQL (system of record, RLS), object store + WORM Object-Lock, Apache Spark/DuckDB (ER scale)	Open-source
Graph	Neo4j + GDS (sovereign alt: Apache AGE on Postgres); Memgraph optional real-time tier	Open-core
Entity resolution	Splink (Fellegi-Sunter + EM); Sentence-BERT + FAISS blocking	Open-source
Analytics / stats	PySAL esda (Gi*, LISA), pymannkendall (EHSA), ST-DBSCAN/HDBSCAN, SaTScan	Open-source
ML / predictive	RTM, LightGBM/CatBoost + SHAP, Hawkes (tick), MAPIE conformal; PyTorch Geometric Temporal (Phase-2)	Open-source
Fairness / MLOps	AIF360, Fairlearn, MLflow, Evidently, DVC, Model Cards	Open-source
AI assistant	Claude (claude-opus-4-8 / Sonnet) in-region + on-prem Llama 3.x/Mistral (vLLM); Mem0; pgvector/Qdrant; Presidio; Llama Guard 4 + Prompt Guard 2	Hybrid
Visualization	MapLibre GL, deck.gl 9, Apache ECharts, Observable Plot, sigma.js v3 + graphology, Cytoscape.js, kepler.gl/pydeck, Gephi	Open-source
Identity & eKYC	FIDO2/passkeys, PKI smartcards, RBAC+ABAC; DigiLocker/Aadhaar eKYC (citizen)	Standards
Security & keys	HSM (FIPS 140-3 L3), HashiCorp Vault (Shamir quorum), Wazuh, Falco, SIEM, Sigstore/SLSA, hash-chained ledger (Trillian-style)	Open-core
Data quality	Great Expectations / Soda, OpenLineage, explicit UNKNOWN states	Open-source
Hosting	On-prem / NIC MeghRaj GovCloud, India region; containerised, CIS-hardened	Sovereign

Why these specific choices are compatible

The stack converges on **PostgreSQL as the gravitational centre** — PostGIS for space, h3-pg for hexagons, pgvector for AI memory, Apache AGE as a sovereign graph option, and Row-Level Security for isolation all live in one hardened, backed-up, audited engine. That minimises the number of systems to secure and keeps the whole platform coherent rather than a loose federation of services.

IX Visualization & Experience Design

Extremely good, not basic — and usable by a constable or a commissioner

Quality bar

- **3D & animated:** deck.gl GPU hex extrusions, animated time-slider “trips”, smooth heat layers — millions of points rendered in-browser.
- **Relational storytelling:** sigma.js WebGL graph canvas with expand-on-click, timeline, and geo-sync.
- **Emerging-trend signalling:** the “red-zone pulsing” alert, grounded in Emerging Hot Spot Analysis — visual urgency backed by real statistics.
- **Uncertainty made visible:** shaded confidence bands and “imputed” badges so honesty is part of the aesthetic, not hidden.

Usability bar

- **Role-tailored views:** a station officer, an SCRB analyst and a commissioner each see an interface pitched at their decisions.
- **Ask-in-plain-language:** the DRISHTI Assistant turns “show vehicle thefts near Majestic after 9pm last month” into a grounded, cited map — no query language needed.
- **Multilingual:** Kannada & English throughout, including MO text matching.
- **Drill-down everywhere:** state → district → station → hex, one consistent interaction model.

Usable *and* uncrackable are not in tension

Ease of use lives in the presentation layer; security lives in the identity, data and key layers beneath it. A clean, plain-language interface sits on top of Zero-Trust access, field encryption and a tamper-evident ledger — the officer gets simplicity, the attacker gets ciphertext.

X Implementation Roadmap

Incremental, value-first, security-first — indicative phasing

PHASE 0 · FOUNDATION

Sovereign platform & security spine

Stand up PostgreSQL/PostGIS + H3, identity (Zero-Trust, RBAC+ABAC), HSM/Vault key governance, the tamper-evident ledger, WORM backups, data-quality & lineage pipeline. *Security and ingestion before any feature.*

PHASE 1 · SEE

Geospatial & hotspot intelligence (M1)

District/station drill-down maps, Gi*/LISA hotspots, Emerging Hot Spot Analysis alerts. First visible win for SCRB; all production-ready methods.

PHASE 2 · CONNECT

Entity resolution & network analysis (M2)

Splink ER with human-review queue, Neo4j graph, Leiden communities, MO/Jaccard serial linkage, sigma.js investigator canvas.

PHASE 3 · UNDERSTAND

Predictive, anomaly & socio-economic (M3, M4)

RTM backbone + LightGBM/SHAP + conformal uncertainty; SaTScan anomaly clusters; GWR context overlays. Bias-audit gate mandatory before go-live.

PHASE 4 · CONVERSE

DRISHTI Assistant (M5) & data engine hardening (M7)

Grounded citation-only assistant with private Mem0 memory, RLS isolation, Presidio redaction, deterministic validator.

PHASE 5 · ENGAGE

MyShield citizen portal (M6)

DigiLocker eKYC, own-records + aggregate stats, DPDP consent, rate-limited and audited. Public-facing — shipped last, after the perimeter is proven.

PHASE 6 · ENHANCE

Experimental accuracy layers (pilots)

Graph WaveNet / GNN link-prediction leads — human-reviewed, explainable, never autonomous; promoted only after monitoring proves them safe.

Sequencing principle

Security and data integrity are built *first* (Phase 0), public exposure comes *last* (Phase 5), and experimental ML is gated behind monitoring (Phase 6). This is the inverse of a hackathon's "demo first, secure later" — and it is the correct order for a police system.

XI Risk Register

Named honestly, with mitigations — because hiding risk from the police is itself a risk

Risk	Mitigation
Predictive-policing bias / feedback loops	Place-based only; predict reported-victimisation categories; AIF360/Fairlearn audits; human-in-the-loop; no automated enforcement.
False links / wrongful implication	Calibrated ER probabilities + review band; explainable links; GNN leads pilot-only and human-reviewed.
AI hallucination	RAG-with-citations, deterministic uncited-claim validator, “data not available” behaviour, conformal uncertainty.
Police-station boundary data unavailable	H3 hex aggregation needs no boundary files; co-develop beat polygons with KSP over time.
Data sovereignty vs Claude cloud	On-prem Llama/Mistral for sensitive raw data; Claude restricted to redacted/aggregate; confirm scope with KSP legal.
Insider exfiltration	ABAC least-privilege, threshold keys, tokenisation, UEBA/DLP, immutable audit ledger.
Ransomware	WORM Object-Lock, air-gapped immutable backups, verified restores — no ransom leverage.
Imputed data mistaken for fact	Physical separation of imputed values, mandatory flags, excluded from operational/evidentiary views.
NeMo Guardrails beta maturity	Used for orchestration only; Llama Guard + deterministic validator are the real safety layers.
Citizen portal as attack surface	eKYC gate, own-data-only queries, rate limiting, anomaly detection, shipped last after perimeter hardening.

XII Why This Is Not a Typical Hackathon Project

The difference between a demo and a system the police can trust

A typical hackathon build	DRISHTI
Pretty charts on a single CSV	Governed platform with validation, lineage, entity resolution and provenance
"AI" that confidently makes things up	Citation-only, deterministically validated, "data not available" by design
Predicts "risky people"	Predicts risky *places & environments*, bias-audited, human-in-the-loop
Security = a login page	8-layer Zero-Trust, HSM keys, tamper-evident ledger, crypto-shred, air-gapped backups
"Unhackable" marketing claim	Honest defence-in-depth: stolen data useless, tampering detectable & reversible
Ignores the law	DPDP, BSA §63 admissibility, CERT-In, data localisation built in

What the parallel-agent research process added

This proposal was assembled by five specialist research tracks working in parallel and cross-checking each other. That process did three things a single pass cannot:

- **Caught dangerous ideas.** The security track explicitly overruled the original "auto-destruct on intrusion" ask — replacing it with crypto-shred + tamper-evidence + immutable backups, which achieves the goal without letting an attacker weaponise the kill-switch.
- **Feasibility-gated every feature.** Cutting-edge methods from current research (EHSA, Hawkes, RTM, conformal prediction, PSI, confidential computing) were included only where verified workable at Karnataka scale; experimental ones are explicitly pilot-only.
- **Made fairness and zero-fabrication structural,** not slogans — enforced in schema, in deterministic validators, and in human-in-the-loop workflow.

The promise to KSP

A platform that is genuinely powerful, genuinely beautiful, genuinely usable — and that earns the one thing a police intelligence system cannot do without: **trust**.

A Appendix — Research Basis & References

Selected sources verified during the parallel research process

Geospatial & spatiotemporal

Esri — **Emerging Hot Spot Analysis** (space-time cube + Mann-Kendall). · Uber **H3** hierarchical hex index (h3geo.org; h3-pg v4.2). · **deck.gl 9** H3HexagonLayer / TripsLayer. · Birant & Kut, **ST-DBSCAN** (2007). · Getis-Ord **Gi*** & Anselin **LISA** via PySAL `esda`. · Hu, Wang, Guin & Zhu — **STKDE / ProMap** with PAI validation (arXiv 2006.00272). · Vector-tile server benchmark 2024 (**Martin**).

Network science & entity resolution

Splink (MoJ) — Fellegi-Sunter probabilistic linkage. · Traag, Waltman & van Eck — **Leiden** algorithm (Sci. Reports 2019; arXiv 1810.08473). · Morselli, **Inside Criminal Networks** (Springer 2009). · Hamilton et al. **GraphSAGE** (NeurIPS 2017); Veličković et al. **GAT** (ICLR 2018). · Bennell & Canter — MO linkage / ROC. · **sigma.js v3** + graphology; Cytoscape.js.

Predictive ML, anomaly & fairness

Caplan & Kennedy — **Risk Terrain Modeling** (UC Press 2016). · Mohler et al. — **self-exciting point process** (JASA 2011). · Wu et al. — **Graph WaveNet** (IJCAI 2019). · Lundberg & Lee — **SHAP** (NeurIPS 2017). · **Conformal prediction** (MAPIE; arXiv 2401.13744). · **AIF360** (IBM), **Fairlearn** (Microsoft), **Evidently**, **MLflow**. · Critiques: RAND / Brennan Center / EFF on Chicago Heat List & PredPol feedback loops.

Security & compliance

NIST SP 800-207 Zero Trust Architecture. · **NIST SP 800-88** Rev.1 (crypto-shred / Purge). · S3 **Object Lock** Compliance Mode (WORM). · AWS **QLDB** Merkle verification (pattern for on-prem Trillian-style ledger). · Confidential computing (Intel TDX/SGX, AMD SEV-SNP, AWS Nitro). · **BSA §63** (Bharatiya Sakshya Adhiniyam 2023) hash certificates. · **CERT-In** Directions 70B (2022). · **DPDP Act 2023 + Rules 2025**. · RFC 3161 timestamping; Shamir Secret Sharing (Vault).

Private assistant, citizen self-check & data quality

Mem0 / Letta (MemGPT) memory frameworks. · **pgvector**, **Qdrant** vector stores. · PostgreSQL **Row-Level Security**. · Microsoft **Presidio** PII redaction. · **Llama Guard 4**, **Prompt Guard 2**, NeMo Guardrails. · **Private Set Intersection** (labeled PSI, IEEE TIFS 2025). · **DigiLocker / Aadhaar eKYC**. · **MICE** multiple imputation; **Great Expectations** / Soda; **OpenLineage**. · Claude model ids (`claude-opus-4-8`) via Anthropic API / Amazon Bedrock.

DRISHTI — Karnataka State Crime Intelligence & Analytical Platform. Proposal v1.0, 14 June 2026. Design & architecture document only; no implementation is included. Every technology named carries a feasibility verdict verified during research; experimental components are explicitly restricted to human-reviewed pilots. Prepared for evaluation by the State Crime Records Bureau, Karnataka State Police.